7.3.1 Conventional suspension system

In this type, wheels are mounted on the two sides of the axle. Leaf spring or coil spring is mounted in between the chassis frame and axle. One end of the leaf spring is attached rigidly usually the front, and the other end is attached through a shackle as movable. The vibration is absorbed by the compression and expansion of leaf spring while travelling over a road with bump and pit. Two ends of the master leaf are connected with the kingpin and knuckle, so that vibration transferred from one side to another. These types of suspension systems are mostly used in rear wheels alone.

Advantages

1. Simple in design
2. Low cost
3. Less number of components
4. Less maintenance

Disadvantages

1. Road shocks from one wheel is transmitted to another wheel. If the road is irregular, the whole vehicle leans on one side.
2. As both the wheels do not get up or down simultaneously, so they will rotate in different positions.

Note

Due to the gyroscopic effect during turning, wheel wobble or wheel shimmy takes place in a wheel. This is a very dangerous problem.

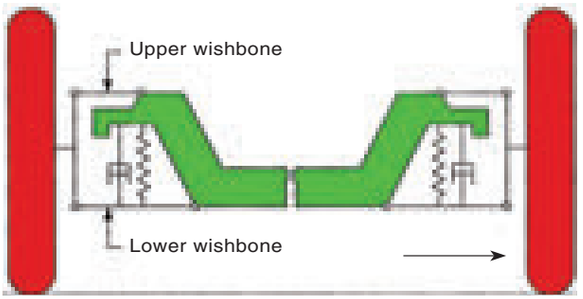
7.3.2 Independent Suspension system

Independent Suspension systems are commonly used in the front axle. When a wheel climbs a bump, the other wheel will travel in its own path. Hence, the vibration will not be transmitted from one wheel to another which leads to smooth ride to the passenger.

Types of Independent front suspension system

1. Wishbone or Parallel link type.
2. Vertical guide type
3. Trailing link type
4. Macpherson strut type
5. Swing half axle type.

7.3.2.1 Wishbone suspension system



Wishbone type independent suspension with coil spring

Figure 7.3.2.1 Wishbone suspension system

This system is shown in the figure 7.3.2.1. This type of suspension system is often used for front wheels. In this system, two arms namely upper wishbone arm and lower wishbone arm are used. The upper wishbone has a short arm and lower wishbone is a long arm and it is connected with the frame. Vibration damper and coil spring are fixed over the rubber pad in between the lower wishbone arm and fixed cross member.

Wishbone Arms keep the wheel in its position and transmit the load to the spring. Therefore, the tractive force,

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